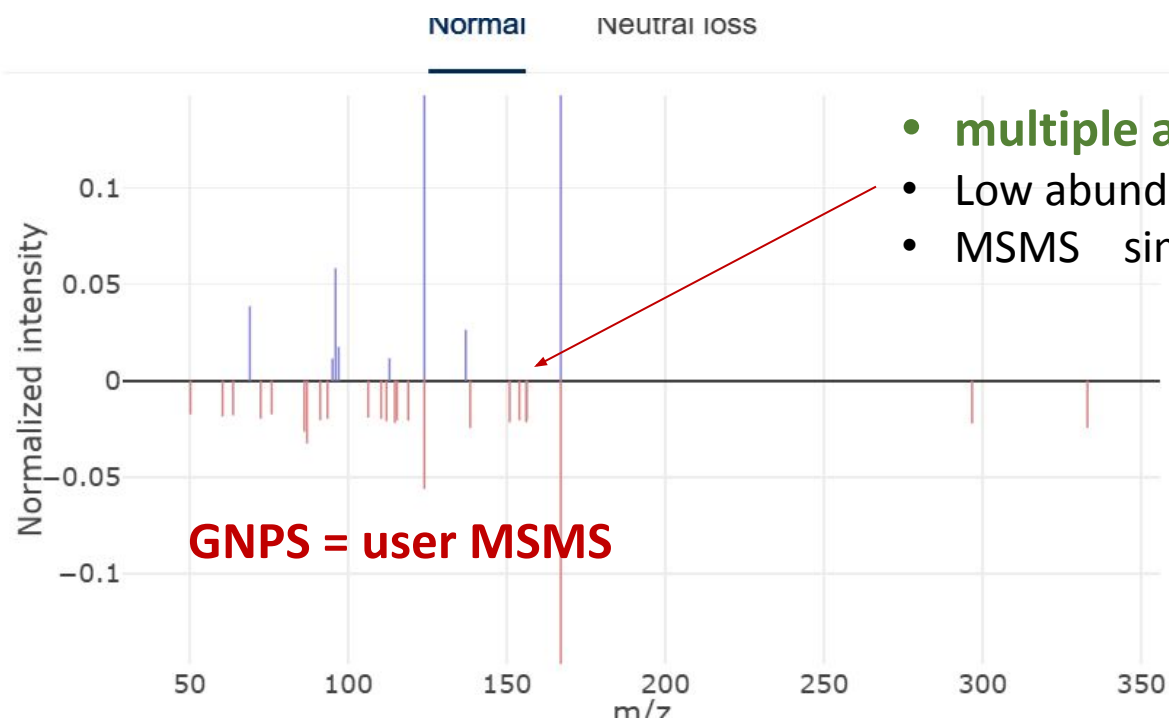


Low Ziyue confidence id 1612, but still ok by Fiehn

112.43	335.049	0.7	uric acid	[2M-H]-	0.961
112.28	357.031	1.0	uric acid	[2M-2H...	0.475
112.76	167.021	1.3	uric acid	[M-H]-	0.901



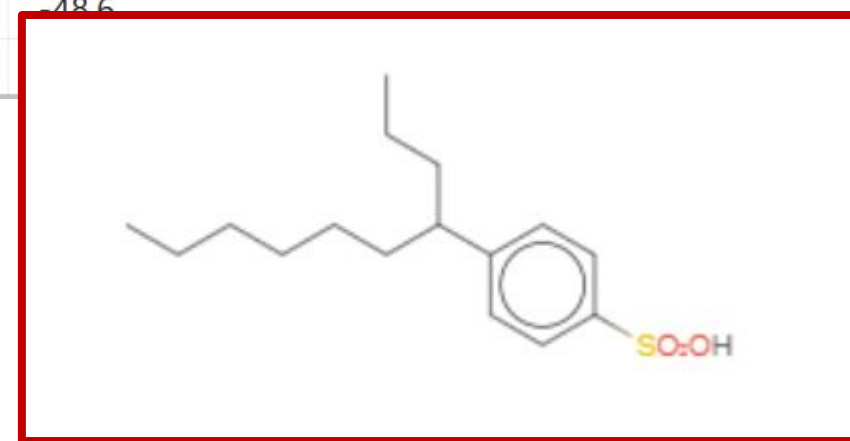
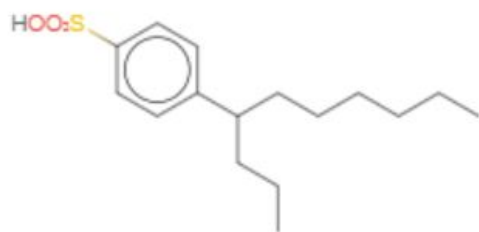
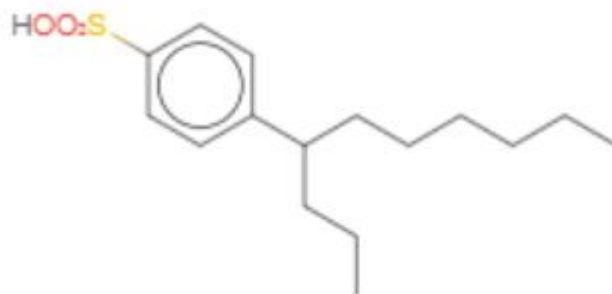
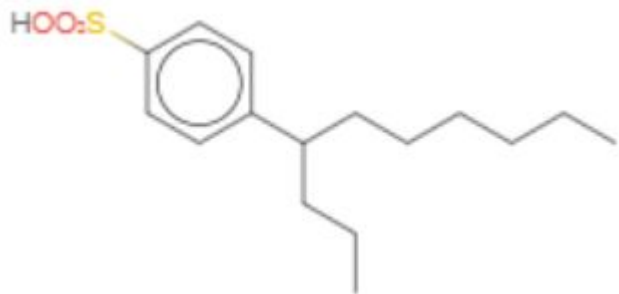
- multiple adducts for uric acid at 112s !
- Low abundant ions are **noise** for both library and experim.
- MSMS sim_gap is **0.368** that's a lot

GNPS	URATE	2M-2H+...	0.475	-0.000	-20.5
Mass...	URATE	[2M-2H...	0.475	-0.000	-20.5
GNPS	Massbank:EQ311856 Triflu...	M-H	0.108	-0.005	-90.2

Low Ziyue confidence id ₂₁₆₀ , but still keep-for-now by Fiehn

Ziyue: RT off (-49s); near-tie (gap=0.029) ?

DB	Name	Adduct	Entropy similarity	Δ mass	Δ predicted RT
Mass...	C10-LAS (STANDARD MIX)	[M-H]-	0.748	-0.001	-48.6
GNPS	MassbankEU:ETS00012 C10-LAS (...)	[M-H]-	0.748	-0.001	-48.6
MB-EU	C10-LAS (STANDARD MIX)	[M-H]-	0.748	-0.001	-48.6
GNPS	MassbankEU:ETS00013 C10-LAS (...)	[M-H]-	0.734	-0.001	-48.6
Mass...	C10-LAS (SAMPLE)	[M-H]-	0.734	-0.001	-48.6
MB-EU	C10-LAS (SAMPLE)	[M-H]-	0.734	-0.001	-48.6



InChIKey

NANHIUZYPFDGJS-UHFFFAOYSA-N

NANHIUZYPFDGJS-UHFFFAOYSA-N

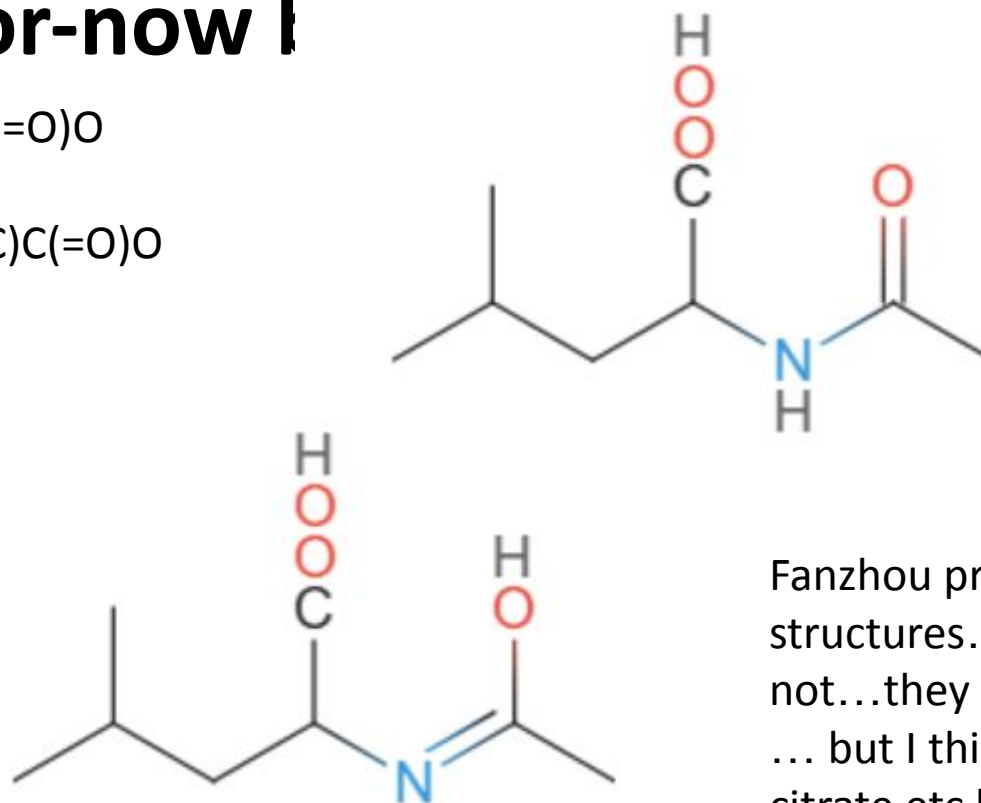
NANHIUZYPFDGJS-UHFFFAOYSA-N

NANHIUZYPFDGJS-UHFFFAOYSA-N

Low Ziyue confidence for id 3281, but still keep-for-now !

CC(=O)NC(CC(C)C)C(=O)O

CC(O)=N[CH](CC(C)C)C(=O)O



Fanzhou predicts identical RT for both structures...why? (-45.7s you could also ask, why not...they are highly similar)
... but I think vicinal acid/alcohols like malate, citrate etc have a lot of retention...

All the N-acetyl-amino acids have very wide spanning pred.RTs and even wide spanning 'experimental' RTs

Need to check with standards

Low Ziyue confidence for id ₂₂₄₄ , but still keep-for-now by Fiehn

C=C(OP(=O)(O)O)C(=O)O

C=C(OP(=O)(O)O)C(=O)O

C=C(C(=O)O)OP(=O)(O)O

- RT off (-39s); ok
- near-tie (gap=0.000); **no....where did Ziyue see that**
- very few peaks (entropy=0.22), **true....but where is that in your formula / algorithm?**

SMILES	InChIKey	Formula
<chem>C=C(OP(=O)(O)O)C(=O)O</chem>	DTBNBXWJWCWCIK-UHFFFAOYSA-N	C3H5O6P
<chem>C=C(OP(=O)(O)O)C(=O)O</chem>	DTBNBXWJWCWCIK-UHFFFAOYSA-N	C3H5O6P
<chem>C=C(C(=O)O)OP(=O)(O)O</chem>	DTBNBXWJWCWCIK-UHFFFAOYSA-N	C3H5O6P

Low Ziyue confidence, but still ok-for-now by Fiehn

MS/MS library

Identity search Open search Neutral loss search Hybrid search							Export	
DB	Name	Adduct	Entropy similarity	Δ mass	Δ predicted RT			
GNPS	N-acetyl-beta-D-galactosa...	Unknown	0.793	0.000	-36.8		Assign molecule	
NIST...	NP-023453	[M+H-C...	0.790	-0.000	-62.9			

RT off (-37s); near-tie (gap=0.003)

- a) We should compare differences in total confidence, not only the near-tie gap for the top 2 hits
- b) Sugars elute surprisingly late...not completely stupid
- (c) Acetyl-amino groups have a strange behavior in Fanzhou's model...and hence, I don't trust the delta RT
- d) What I really don't like is "adduct unknown"...we should also check adduct and accurate mass

Monoisotopic Mass 221.090

Mz experimental 204.087

= delta M 17.003

= [M+H-H₂O]⁺ = ISF = 17.003

Resubmitted to LCB ...but then checked others entries in LCB for this compound ...next slide

Low Ziyue confidence, and Fiehn *now agrees* after checking other LCB entries for *that bin...and looking at the structure*

▼ Spectra list

5m hilic premier | orbitrap | beh amide | positi

ListPlot

Clear sorting (1)

Name

▼

Q galactosamin

✕

7 results

☒

Show spectra without MS/MS peaks

⬇

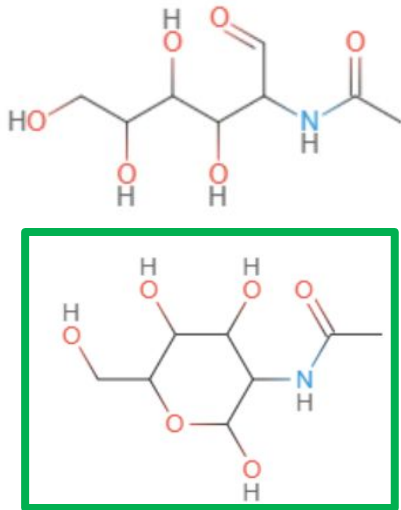
Export

⚙

	ID	RT	m/z	Entropy	Name	Adduct	Identity Score	Fuzzy Score	SMILES
1	9179	149.93	608.089	2.8	UDP-N-acetyl-D-galactosamine	[M+H] ⁺	0.879	0.904	CC(=O)NC1C(C(C(OC1OP(=O)(O)OP
2	5497	139.81	204.087	3.0	N-acetyl-galactosamine	[M+H-H2O] ⁺	0.793	0.794	CC(=O)NC(C=O)C(O)C(O)C(O)CO
3	6602	52.29	204.087	2.7	N-acetyl-galactosamine	[M+H-H2O] ⁺	0.985	0.985	CC(=O)NC1C(C(C(OC1O)CO)O)O
4	7543	129.53	162.076	2.8	galactosamine	[M+H-H2O] ⁺	0.872	0.872	C(C(C(C(C(C=O)N)O)O)O)O
5	9531	42.91	204.087	2.3	yy_N-acetylgalactosamine_delta RTpred +3...	[M+H-H2O] ⁺	0.980	0.980	CC(=O)N[CH]1[CH](O)O[CH](CO)[CH
6	6603	52.29	186.076	3.1	N-acetyl-galactosamine	[M+H-2H2O] ⁺	0.949	0.949	CC(=O)NC1C(C(C(OC1O)CO)O)O
7	10679	101.59	204.087	2.4	N-acetyl-galactosamine	Unknown	0.714	0.848	CC(=O)NC(C=O)C(O)C(O)C(O)CO

But still unclear. **No non-ISF entry.** And entries at 52s are +24s off (i.e. predRT 76s),
whereas entry id at 102s is +3s off = 105s □ **open chain/ closed cycle...**

RT	m/z	Entropy	Name	Adduct	Identity Score
139.81	204.087	3.0	yy_N-acetyl-galactosamine	[M+H-H ₂ O] ⁺	0.793
101.59	204.087	2.4	N-acetyl-galactosamine	[M+H-H ₂ O] ⁺	0.714
52.29	204.087	2.7	N-acetyl-galactosamine	[M+H-H ₂ O] ⁺	0.985
52.29	186.076	3.1	N-acetyl-galactosamine	[M+H-2H ₂ O] ⁺	0.949
42.91	204.087	2.3	yy_N-acetyl-galactosamine	[M+H-H ₂ O] ⁺	0.980



Low Ziyue confidence, but still ok-for now by Fiehn

MS/MS library

Identity search

Open search

Neutral loss search

Hybrid search

↓ E

DB	Name	Adduct	Entropy similarity	Δ mass	Δ predicted RT
NIST...	Trigonelline cation	[Cat]+	0.989	-0.000	-46.6
NIST...	Trigonelline cation	[Cat]+	0.982	-0.000	-46.6
GNPS	Trigonelline	M+H	0.976	-0.000	-9.4
NIST...	Trigonelline	[M+H]+	0.972	-0.000	-9.4
NIST...	Trigonelline	[M+H]+	0.972	-0.000	-9.4
NIST...	Trigonelline cation	[Cat]+	0.954	-0.000	-46.6

ID	RT	m/z	Entropy	Name	Adduct	Identity Score	Fuzzy Score	SMILES
1674	106.48	275.102	0.7	trigonelline	[2M+H]+	0.998	0.998	<chem>C[N+]1=CC=CC(=C1)C(=O)[O-]</chem>
10123	106.02	138.055	1.8	trigonelline	[M+H]+	0.989	0.989	<chem>C[N+]1=CC=CC(=C1)C(=O)[O-]</chem>
7585	106.44	297.084	0.8	trigonelline	[2M+Na]+	0.905	0.905	<chem>C[N+]1=CC=CC(=C1)C(=O)[O-]</chem>
4156	50.47	138.091	1.8	yy_trigonelline_36 mDa error		0.603	0.855	
3029	106.07	160.037	1.2	trigonelline	M+Na	0.582	0.731	<chem>C[n+]1cccc(C(=O)[O-])c1</chem>

Low Ziyue confidence, but still ok-for-now by Fiehn

▼ MS/MS library

Identity search

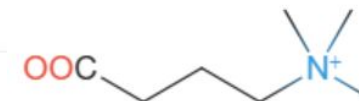
Open search

Neutral loss search

Hybrid search

↓ Exp

DB	Name	Adduct	Entropy similarity	Δ mass	Δ predicted RT
NIST...	(3-Carboxypropyl)trimethy...	[Cat]+	0.854	0.000	-37.2
NIST...	(3-Carboxypropyl)trimethy...	[Cat]+	0.853	0.000	-37.2
GNPS	Deoxycarnitine (known str...	[M+H]+	0.852	-0.001	-7.1
NIST...	(3-Carboxypropyl)trimethy...	[Cat]+	0.828	0.000	-37.2
NIST...	(3-Carboxypropyl)trimethy...	[Cat]+	0.828	0.000	-37.2
GNPS	DEOXYCARNITINE	M+H	0.828	0.000	-7.1



Low Ziyue confidence, but still ok by Fiehn